

LA CROSSE, WI, USA

WMO: 726430

Lat: 43.879N

Lon: 91.253W

Elev: 199

StdP: 98.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.3	-27.6	0.3	-22.1	-25.1	0.4	-19.9	11.3	-7.1	10.2	-7.6	2.6	340	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.0	23.9	31.4	22.8	30.0	21.9	25.4	30.9	24.3	29.5	23.3	28.3	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.0	28.8	22.5	17.7	27.5	21.6	16.7	26.5	78.8	30.6	74.2	29.7	70.2	28.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	8.0	DB	-26.8	36.2	4.3	2.2	-29.9	37.8	-32.4	39.1	-34.8	40.4	-37.9	42.0	(4)
(5)			WB	-27.1	27.1	4.1	1.3	-30.1	28.1	-32.5	28.8	-34.8	29.6	-37.8	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-7.6	-5.3	1.7	9.2	15.7	21.2	23.6	22.2	18.2	10.8	3.0	-4.2	(6)	
	DBStd	12.10	7.13	6.89	6.66	5.38	4.66	3.73	3.14	3.10	4.48	5.03	5.53	6.45	(7)	
	HDD10.0	2042	547	429	269	78	7	0	0	0	2	51	217	442	(8)	
	HDD18.3	3849	806	663	517	278	109	16	2	4	57	239	459	699	(9)	
	CDD10.0	1716	0	0	12	53	183	337	421	379	247	75	8	1	(10)	
	CDD18.3	481	0	0	1	3	27	102	165	125	53	6	0	0	(11)	
	CDH23.3	4258	0	0	5	42	276	902	1529	1022	432	50	0	0	(12)	
	CDH26.7	1451	0	0	0	9	81	313	587	323	129	9	0	0	(13)	
(14) Wind	WSAvg	3.8	3.8	3.9	3.9	4.3	3.9	3.5	3.3	3.1	3.5	3.9	4.1	3.8	(14)	
(15) Precipitation	PrecAvg	761	26	23	49	77	97	103	90	92	96	53	39	27	(15)	
	PrecMax	1120	73	56	94	186	247	242	233	250	267	112	88	65	(16)	
	PrecMin	444	5	1	8	15	24	34	4	14	14	6	0	8	(17)	
	PrecStd	154	19	16	23	42	53	56	51	61	61	30	26	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	12.9	22.9	28.0	32.2	34.6	36.3	33.9	32.5	27.9	20.2	12.9	(19)	
		MCWB	5.1	9.2	15.8	17.2	21.4	23.9	25.6	24.9	22.7	19.7	13.9	10.4	(20)	
	2%	DB	5.4	9.1	18.1	23.6	28.9	32.1	33.4	31.8	30.0	24.3	16.6	8.1	(21)	
		MCWB	2.8	5.7	12.3	14.1	19.2	22.9	24.7	23.5	22.0	17.2	12.1	5.5	(22)	
	5%	DB	3.1	6.0	14.6	21.0	26.4	30.1	31.5	30.0	27.9	21.5	13.6	5.8	(23)	
		MCWB	1.4	3.2	9.9	12.4	17.8	21.5	23.3	22.4	20.5	15.9	9.5	3.7	(24)	
	10%	DB	1.6	3.6	11.1	18.3	24.0	28.2	29.9	28.3	25.9	18.9	11.1	3.7	(25)	
		MCWB	0.3	1.5	7.1	10.8	16.5	20.6	22.3	21.5	19.3	14.2	7.5	1.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	9.1	16.9	18.4	23.1	25.5	27.5	26.4	24.5	21.5	14.9	11.0	(27)	
		MCDB	8.2	13.0	21.9	25.9	30.2	32.1	33.0	31.9	30.3	26.1	18.6	12.4	(28)	
	2%	WB	3.1	5.8	13.9	15.5	20.8	24.1	25.8	24.9	22.9	18.8	12.7	6.0	(29)	
		MCDB	4.9	8.8	16.9	21.7	26.5	30.2	31.5	29.5	27.8	22.7	16.0	7.3	(30)	
	5%	WB	1.6	3.5	10.2	13.6	19.3	22.9	24.6	23.8	21.7	16.6	10.1	3.7	(31)	
		MCDB	3.1	5.9	14.3	19.1	24.3	28.7	30.0	28.2	26.2	20.6	13.2	5.6	(32)	
	10%	WB	0.4	1.7	7.0	11.6	17.7	21.6	23.5	22.7	20.5	14.7	7.8	1.9	(33)	
		MCDB	1.6	3.4	10.5	17.4	22.6	26.6	28.4	27.0	24.6	18.4	10.8	3.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	9.9	10.9	11.2	10.7	10.8	10.5	11.1	9.9	8.2	7.4	(35)	
		MCDBR	9.1	10.1	13.8	15.3	14.2	12.7	12.3	11.8	12.9	13.3	11.9	9.0	(36)	
	5% WB	MCWBR	7.3	7.3	8.2	8.0	6.6	5.5	5.2	5.0	6.0	7.4	7.9	7.1	(37)	
		MCWBR	8.8	9.8	12.7	13.0	12.2	11.3	10.9	10.4	11.1	11.5	11.1	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.266	0.296	0.327	0.367	0.397	0.408	0.412	0.406	0.373	0.332	0.297	0.269	(40)	
		taud	2.391	2.314	2.375	2.326	2.294	2.297	2.291	2.318	2.392	2.496	2.511	2.444	(41)	
	Ebn at Noon	Ebn at Noon	872	897	912	896	875	863	855	848	851	849	831	835	(42)	
		Edh at Noon	75	97	106	121	128	129	128	120	103	82	68	65	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.78	2.85	3.88	4.53	5.36	5.95	6.16	5.23	4.20	2.69	1.76	1.30	(44)	
	RadStd	RadStd	0.18	0.28	0.45	0.44	0.41	0.46	0.29	0.35	0.31	0.36	0.15	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page